Roll No. ....

TEE–702(NCER)

B. TECH. (EE) (SEVEN SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022

NON-CONVENTIONAL RESOURCES AND ENERGY SYSTEMS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What are conventional and non-conventional energy resources ? Write the advantages and limitations of non-conventional energy resources. 10 Marks (CO1, CO2)

OR

(b) Define declination angle, hour angle, zenith angle, solar azimuth angle and angle of incidence. 10 Marks (CO1, CO2)

2. (a) Explain the I-V characteristics of solar cell and define fill factor. What is the significance of fill factor ? 10 Marks (CO1, CO2)

OR

(b) Explain the MPPT in an SPV system. Name the various strategies used for operation of an MPPT. 10 Marks (CO1, CO2)

3. (a) With the help of block diagram, explain the operation of standalone and grid interactive solar generation plant. 10 Marks (CO1, CO2)

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OR

(b) What do you understand by energy payback period ? Explain it in terms of photovoltaic generation ?

10 Marks (CO1, CO2)

4. (a) With the help of a schematic diagram, explain solar passive space cooling system through ventilation.

10 Marks (CO1, CO2)

OR

(b) Explain the depletion process of solar radiation as it passes through the atmosphere to reach at the surface of the earth.

10 Marks (CO1, CO2)

5. (a) With the help of a schematic diagram, explain the working of solar water heating.

10 Marks (CO1, CO2)

OR

(b) Explain the mechanism of photoconduction in a photovoltaic cell.

10 Marks (CO1, CO2)